Diagnostic Methods for Malassezia Folliculitis

Calcofluor white/Blancophor staining, which requires fluorescence microscopy, has been used for the detection of *Malassezia*. The diagnostic performance of Gram staining was evaluated in 26 patients with *Malassezia* folliculitis (MF), diagnosed based on at least two of the following criteria: (a) typical clinical presentation, (b) histopathological identification of *Malassezia* within inflamed hair follicles, or (c) clinical response to antifungal therapy.[20] Gram staining correctly identified 22 out of 26 patients, yielding a sensitivity of 84.6%. In the same study, all six patients without MF were correctly classified as negative, resulting in a specificity of 100%.[20]

The diagnostic accuracy of potassium hydroxide (KOH) preparation and May-Grünwald-Giemsa (MGG) staining was assessed in 49 patients with MF confirmed by cytological examination, fungal culture, and molecular analysis.[7] KOH correctly identified 40 patients (sensitivity 81.6%), while MGG detected all 49 cases (sensitivity 100%). Since all participants in this study had MF, specificity for KOH and MGG could not be determined.[7]

Histopathological Examination

For histopathological evaluation, a punch biopsy is typically obtained from the central area of a follicular lesion and stained with haematoxylin-eosin (H&E) or periodic acid–Schiff (PAS). Characteristic findings include dilated and plugged hair follicles associated with a mixed inflammatory infiltrate composed of eosinophils, lymphocytes, neutrophils, and plasma cells.[6,21]

Molecular-Based Detection

Molecular techniques such as polymerase chain reaction (PCR) and matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS) can identify specific *Malassezia* species and genotypes, thereby supporting the diagnosis of MF.[6,19,22–24] However, because *Malassezia* is part of the normal cutaneous microbiota, molecular detection alone cannot confirm pathogenic involvement.[6,22] PCR methods used include real-time PCR, end-point PCR, restriction fragment length polymorphism (RFLP)-PCR, terminal RFLP (tRFLP), and PCR sequencing.[6,22] Despite their utility, neither PCR nor MALDI-TOF MS is routinely available in most clinical settings.

Fungal Culture

Fungal cultures can be performed using specialized media such as Dixon's agar, Leeming-Notman agar, or Ushijima's medium.[19,25] Culture results may support the diagnosis of MF and provide antimycotic susceptibility data. However, contamination with non-pathogenic *Malassezia* from adjacent normal skin is a potential pitfall.[6,26] Additionally, growth rates vary among *Malassezia* species, which may skew results, and these culture methods are not widely accessible in routine clinical practice.[6,12]

Wood's Light Examination

During Wood's light examination, ultraviolet light (320–400 nm) is applied to the skin, causing MF lesions to fluoresce with a green-yellow hue. In a study of 49 patients with MF confirmed by cytology, culture, and molecular methods, the sensitivity of Wood's light was 65.3%. While this method can support clinical diagnosis, it lacks genus and species specificity (see Table 1).

Comments on the Diagnosis of Malassezia Folliculitis

Direct microscopic examination (e.g., KOH, MGG) is a non-invasive method with high diagnostic accuracy and is recommended when clinical suspicion of MF exists. However, it is not currently required before initiating topical antifungal therapy. Although this may lead to diagnostic uncertainty, microscopic evaluation is not universally available, and delays in treatment initiation may occur. Topical antifungals have demonstrated cure rates up to 100% in some studies, with minimal adverse effects (Table S5).[28,29] Therefore, in typical cases, the benefits of clinical diagnosis without mandatory microscopy outweigh the risks. Nevertheless, microscopic examination remains a valuable supplementary tool, particularly in refractory or atypical cases.

Treatments for Malassezia Folliculitis

Evidence for MF treatment is limited to one randomized controlled trial evaluating oral itraconazole, along with several non-randomized comparative and non-comparative interventional studies, observational reports, and case series. Treatment recommendations are summarized in Table 2, with supporting evidence detailed in Tables S5 and S6.